

ORIGINAL ARTICLE

Canopy position and tree size structure the distribution of honeydew-producing scale insects (Hemiptera: Coccoidea) in a temperate hardwood forest

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Funding information

Natural Sciences and Engineering Research Council of Canada

Associate Editor: Christopher Hassall

Abstract

1. Forest canopies are vertically stratified habitats in which microclimate and host tissue quality vary with height. Scale insects (Hemiptera: Coccoidea) are ecologically important components of temperate forest canopies, producing honeydew exploited by bees, wasps and lepidopterans. The spatial distribution of tree scale insects thus has direct consequences for the availability of these resources but has not been described in North American hardwood forests.
2. We quantified the abundance of three scale insect species and unidentified crawlers across upper-, middle- and lower-canopy strata of 33 sugar maple (*Acer saccharum* Marsh.) trees in south-central Ontario. We tested whether canopy position, tree size and year-over-year growth predicted scale abundance using per-species negative binomial generalized linear mixed-effects models (GLMMs).
3. Scale insect communities were strongly vertically stratified, with all taxa reaching highest densities in the upper canopy. Species differed in their association with internode age: *Lepidosaphes ulmi* (L.) and crawlers peaked on the current year's leader, while *Parthenolecanium corni* (Bouché) peaked on the prior year's internode, particularly in large-diameter trees. Tree size had no significant main effect for any taxon, though species-specific patterns emerged within canopy strata.
4. Shoot elongation was positively associated with subsequent scale abundance across multiple canopy positions and growth stages, while radial growth was negatively associated with abundance in the upper and mid canopy. After controlling for shared habitat use, *Phenacoccus acericola* King, *Parthenolecanium* and crawlers showed positive co-variation at the branch scale; the crawler pool was most closely associated with *Parthenolecanium*.
5. Within-tree habitat heterogeneity, rather than tree-level traits, drives the spatial structure of scale insect communities, with implications for the distribution of honeydew resources supporting canopy-dependent arthropods.

KEYWORDS

canopy stratification, honeydew, scale insects, within-tree distribution

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INTRODUCTION

The diversity of canopy arthropods is essential for stability, resilience and productivity of forest ecosystems (Ampoorter et al., 2020; Nakamura et al., 2017). Temperate deciduous forests are among the most prevalent worldwide, yet studies that examine canopy arthropod communities are lacking when compared with tropical regions. Within temperate forests, individual trees comprise a vertically stratified environment that varies in abiotic factors such as temperature (Chilima & Leather, 2001; Dale & Frank, 2017; Khaladi et al., 2024), wind exposure (Platková et al., 2020), resource availability (Speight et al., 1998) and biotic factors including exposure to species-specific diseases and parasites (Hanks & Denno, 1993; Patankar et al., 2014; Schowalter, 1989). Though it is apparent that temperate forest canopies are less complex than most forests in tropical regions, it is also clear that many organisms depend on and are vertically stratified in temperate forest canopies (Wardhaugh et al., 2006).

In most cases, arthropods that are sessile, concealed feeders, such as sap-sucking scale insects, stem-borers and leaf-miners, are less strongly affected by structural heterogeneity within their environment compared to surface-active external feeding species (Wardhaugh et al., 2006). However, it is often difficult to obtain accurate habitat-specific population densities of insect herbivores in either group (Lowman & Wittman, 1996). At large spatial scales, factors such as precipitation and altitude can affect insect population densities (Wardhaugh, 2014), while tree species composition, tree size and tree physiological status are believed to be influential at smaller spatial scales (Nock et al., 2008; Wardhaugh, 2014; Wardhaugh et al., 2006). For example, herbivorous insects often exhibit greater fecundity on higher quality hosts (Awmack & Leather, 2002), but heavy infestations can ultimately negatively impact the host, thereby influencing both inter- and intraspecific interactions (Linghu et al., 2023; Smith & Schowalter, 2001). The quality of available resources can also shift as trees mature; for example, mature sugar maples have been shown to offer less favourable leaf traits to herbivores compared to younger trees (Hakimara & Despland, 2025).

Scale insects (suborder Sternorrhyncha, Hemiptera) are common in temperate forests and evidence suggests that honeydew, the sugary waste produced by scale insects, is an important resource for a variety of other arthropods and can otherwise strongly influence arthropod community composition (Ewers, 2002; Morales et al., 2016). A critical aspect of honeydew is that it can provide arthropods with a vital carbohydrate resource (Eubanks & Styrsky, 2005). Scale insects process large quantities of phloem sap to meet their nutritional requirements for nitrogen (Chilima & Leather, 2001; Grant & Beggs, 1989; Mattson, 1980), and as a consequence, honeydew can be found in abundance and is readily available throughout the summer in temperate forests. As indicated by a study in New Zealand, honeydew can be a limiting resource for a variety of arthropods, with potential impacts as well on other organisms such as birds (Ewers, 2002). Hence, it is important to understand scale distribution within tree canopies, as it pertains to the inherent availability of honeydew.

To date, most work on vertical stratification in temperate forest insect communities has focused on either flying insects (Birtele & Hardersen, 2012; Maguire et al., 2014; Smith et al., 2012; Vance et al., 2007) or leaf-feeding herbivores (Phillipson & Thompson, 1983; Corff & Marquis, 1999; Thomas, 2010; Seifert et al., 2020); more recent studies have broadened this picture to include predators and detritivores (Šigut et al., 2018; Xing et al., 2023). Limited literature on patterns of herbivory through tree ontogeny also indicates that there are both large differences in the herbivore community throughout tree development as well as differences in tree defence strategies (Boege et al., 2011). Scale insects feed on the woody parts of plants and thus represent a distinct feeding guild that has received very little attention in terms of vertical stratification or ontogeny effects. Bark thickness is likely to be a main determinant of distribution patterns of sucking scale insects (Jactel et al., 1996; Nicolai, 1986), as is the chemistry of phloem on which the scales feed, and the seasonal phenology of feeding and honeydew excretion that can shape when carbohydrate subsidies are available. Scale insects show adaptations that enable them to better access phloem, and thus large amounts of water (Hanks & Denno, 1993). As a result, they are less likely to be affected by desiccation than leaf feeders such as lepidopteran larvae, which tend to be most abundant in the lower parts of the canopy (Thomas et al., 2010). Several other factors (proximity to carbohydrate sources, higher tissue nitrogen levels, thinner bark) may also predispose scale insects to be distributed in the upper canopy of trees.

Scale insect honeydew represents a spatially structured carbohydrate subsidy that can support bees, parasitic wasps and other canopy-dependent arthropods (Morales et al., 2016), making the within-tree distribution of scale insects ecologically consequential for where these resources become available. Despite their high abundance in temperate forest systems, scale insect distribution patterns remain understudied. Most prior research on scale insect distribution patterns has been conducted in beech forests from the southern hemisphere (Kelly, 1990; Wardhaugh et al., 2006; Wardhaugh & Didham, 2005, 2006); to our knowledge there have been no studies exploring scale distribution patterns in forests of North America. Since trees generally reach maximum photosynthetic capacity and foliar N at intermediate life-history stages (Thomas, 2010), scale abundance might be expected to be highest in medium-sized trees. This is consistent with prior observations of scale abundance on *Nothofagus* in New Zealand (Gaze & Clout, 1983; Kelly, 1990). Since late-instar scale insects are sedentary, this makes them an amenable study organism for examining their spatial distribution within tree crowns. Differences in distribution patterns among scale insect species have also received little attention to date; however, their relative abundances are most likely to be positively correlated because similarities in their biology will tend to dictate similar patterns in habitat suitability (Crozier, 1981; Gaze & Clout, 1983; Kelly, 1990; Wardhaugh & Didham, 2005).

In the present study, we assessed scale insect distribution patterns in relation to tree size and vertical stratification within crowns of sugar maple (*Acer saccharum* Marsh.), a dominant tree species in northern hardwood forests of North America. To survey scale

populations, we counted scale insects on branch samples taken from harvested trees of varying size and at different heights within the crown. Specifically, we tested four hypotheses: (H1) vertical microclimate and host tissue quality within tree crowns structure scale insect distributions such that conditions in the upper crown are more favourable for scale establishment and persistence; accordingly, we predicted scale insect abundance to increase with canopy height, with higher densities in the upper crown. (H2) Host suitability for scale insects changes with tree ontogeny, as bark architecture, vigour and resource allocation shift across size classes, producing a nonlinear relationship between tree diameter and scale abundance; we expected scale abundance to vary nonlinearly with tree size, peaking at intermediate diameter classes. (H3) Co-occurring scale insect species respond similarly to shared host and microhabitat conditions, leading to concordant abundance patterns across trees and crown positions; we therefore expected abundances of different species to be positively correlated across samples. (H4) Branch-level host vigour mediates scale colonization and survival, with prior-year internode growth serving as an indicator of resource availability that shapes subsequent scale abundance; this predicts a positive association between prior-year internode growth and subsequent scale abundance at the branch level.

MATERIALS AND METHODS

Scale insect surveys

Field sampling was conducted in September 2015 at the Haliburton Forest and Wildlife Reserve Ltd. located in the Great Lakes-St. Lawrence forest region of Ontario, Canada (45°15' N, 78°34' W). Sampling occurred within a recently harvested hardwood forest stand managed under selection-system silviculture following guidelines established by the Ontario Ministry of Natural Resources (OMNR, 2015). The harvested stand encompassed approximately 1–2 ha. The stand was uneven-aged, with some large relict trees (>100 years old) and was closed-canopy with an estimated basal area of $\sim 30 \text{ m}^2 \text{ ha}^{-1}$ prior to harvest. Forest inventory data from permanent sample plots established nearby in 2000, indicated a basal area of $20.6 \text{ m}^2 \text{ ha}^{-1}$ and stem density of $595 \text{ stems ha}^{-1}$ ($\geq 10 \text{ cm dbh}$). Composition by basal area was dominated by sugar maple (*Acer saccharum* Marsh., 39%), balsam fir (*Abies balsamea* (L.) Mill., 25%), American beech (*Fagus grandifolia* Ehrh., 14%) and Eastern hemlock (*Tsuga canadensis* (L.) Carrière, 10%), with lesser amounts of yellow birch (*Betula alleghaniensis* Britt), black ash (*Fraxinus nigra* Marshall), red maple (*Acer rubrum* L.), mountain maple (*Acer spicatum* Lam.), striped maple (*Acer pensylvanicum* L.) and basswood (*Tilia americana* L.).

Sampling took place within 3 days of the area being harvested in August 2015, and tree crowns were left on the forest floor to facilitate branch sample collection at various heights within the canopy. A total of 33 sugar maple (*A. saccharum*) trees were sampled, all selected because their felled crowns could be confidently matched to their corresponding trunk stumps. Sampled trees were distributed throughout

the stand and were separated by a minimum of approximately 10 m, reducing the likelihood of repeatedly sampling immediately adjacent individuals. Trees spanning a range of sizes were intentionally chosen to capture variation associated with different ontogenetic stages. Tree diameter at breast height (dbh) was estimated using a locally calibrated linear equation relating to stump diameter (Wharton, 1984). Estimated dbh values ranged from approximately 15 cm to >55 cm. For statistical analysis, trees were subsequently grouped into three size classes (small: 15–30 cm; medium: 30–55 cm; large: >55 cm dbh) to reflect broad ontogenetic differences among trees. We treated dbh as an integrative proxy for tree ontogenetic stage. Larger dbh is associated with greater foliar biomass and habitat volume, more developed (fissured) bark architecture (Gauthier & Guillemette, 2018) and ontogenetic shifts in phloem nitrogen and leaf quality (e.g., Thomas, 2010). We therefore interpreted effects of dbh as the net outcome of these correlated changes in habitat and resource availability. Most sampled trees fell within the mid-size classes, with fewer individuals in the smallest and largest classes (Figure S1); this size structure is characteristic of a selection-managed hardwood stand, in which many of the largest trees are retained at each harvest entry rather than removed.

From each sampled tree, branch material was collected from the upper, middle and lower thirds of the live crown only; five branches were sampled per stratum per tree. Canopy strata position refer exclusively to vertical stratification within the crown, visually divided into three approximately equal vertical sections based on the position of branches along the main stem axis. As precise crown-length measurement was not possible for felled, ground-level crowns; crown sections were therefore assigned proportionally based on branch position relative to the crown base and tip. Scale insects were counted on the main leader branch of each sample for the four most recent years of growth. Annual growth segments were identified as discrete internodes along the leader, with each internode corresponding to 1 year of growth. For each internode, length (cm) was measured using a ruler, and width (mm) was measured using digital callipers. All scale insects present on that internode were counted. Counts associated with a given internode were assigned to the year in which that internode was produced, allowing scale insect abundance to be linked to specific annual growth increments. Individual specimens were identified using a dissecting microscope as: crawlers (immature stage), *Parthenolecanium corni* Bouché (hereafter *Parthenolecanium*), oystershell scale (*Lepidosaphes ulmi* L.) (hereafter *Lepidosaphes*) or maple mealybug (*Phenacoccus acericola* King) (hereafter *Phenacoccus*) using taxonomic keys (Miller & Davidson, 2005). Sample specimens of each species and a mix of crawlers were preserved in 95% alcohol and voucher specimens deposited at the Royal Ontario Museum (Toronto, Canada) at the Canadian National Collection of Insects (Ottawa, Canada).

Statistical analysis

All analyses were conducted in R version 4.5.2 (R Core Team, 2025) and were designed to test the four hypotheses outlined in the

Introduction. To test hypotheses concerning canopy stratification and host tree size effects (H1–H2), per-species generalized linear mixed-effects models (GLMMs) were fit using the *glmmTMB* package (Brooks et al., 2017), with a negative binomial (NB2) error distribution to account for over-dispersion. Because abundance may vary both with fine-scale differences among annual shoot segments and with coarser differences among host trees, we fitted two complementary models per response at different observational scales. At the internode scale, we modelled counts per internode as a function of canopy stratum, internode age and their interaction, with tree identity and branch identity (nested within tree) included as random effects to account for repeated sampling within trees and for the non-independence of the four internodes within each branch. At the branch scale, we summed counts across the four internodes within each branch and modelled branch totals as a function of canopy stratum, tree size and their interaction, with tree identity as a random effect. To evaluate predicted nonlinear relationship between tree size and scale abundance using a continuous measure, we additionally fit a parallel per-species branch-scale model with dbh (including a linear and quadratic term) in place of tree size class. Pairwise differences among estimated marginal means were assessed using Tukey-adjusted contrasts computed with the *emmeans* package (Lenth, 2024), and compact letter displays were generated with the *multcomp* package (Hothorn et al., 2008).

To evaluate whether scale insect taxa co-occur on branches beyond shared habitat associations (H3), we quantified pairwise associations among taxa using Spearman's rank correlations at the branch scale at two levels. First, raw correlations of branch-level scale insect densities (counts summed across the four annual internodes within each branch, divided by total lateral branch surface area: $\pi \times (\text{width}/10) \times \text{length}$, summed across internodes) were computed using the *Hmisc* package (Harrell, 2024). Spearman's rank correlation was selected because the density distributions departed strongly from normality. Second, to assess whether observed associations persisted after accounting for shared habitat preferences, we fit a per-species linear model of log-transformed branch density ($\log(\text{density} + 0.001)$, where 0.001 accommodates zero values) with canopy position and tree size as categorical predictors. Residuals from each model were then Spearman-correlated across taxa, yielding habitat-controlled correlation coefficients that represent co-variation beyond canopy stratum and tree size.

To test whether prior-year branch growth influenced scale insect abundance the following year (H4), we fit a generalized linear mixed-effects model with a negative binomial error distribution and log link to account for over-dispersed count data. The response variable was the number of scale insects counted on each annual internode of a sampled branch. Because each internode represents one full year of branch elongation, we indexed internodes sequentially outward from the current year's leader (internode 1) through progressively older segments (internodes 2, 3 and 4). Branch growth was quantified at the internode level using two metrics: internode length (annual elongation) and internode width (annual radial growth measured at the internode midpoint). For each scale count on

internode N , we computed the year-over-year change in each growth metric as the difference between internode N and the next older internode ($N + 1$): $\Delta\text{length} = \text{length}(N) - \text{length}(N + 1)$, and analogously for width. Positive values indicate accelerating branch growth (year N grew more than year $N + 1$), and negative values indicate decelerating growth. We refer to each focal–older pairing as a “growth stage” labelled by the older–younger interval: stage 2–1 pairs the change in growth between internodes 2 and 1 with the scale count on internode 1, stage 3–2 pairs the change between internodes 3 and 2 with the count on internode 2 and stage 4–3 pairs the change between internodes 4 and 3 with the count on internode 3. Fixed effects were growth change (Δlength or Δwidth), growth stage and canopy stratum (upper, mid, lower), with the full three-way interaction to allow growth–abundance slopes to vary by stratum and stage.

Separate models were fit for elongation and radial growth metrics. Prior to fitting the radial growth model, five observations with implausible year-over-year width changes ($|\Delta\text{width}| > 20$ mm) were identified as data entry errors and excluded; all other analyses retained the full dataset. Tree size class was initially included but was removed from final models after it did not alter growth-stage- or canopy-position-specific slope estimates. Inference focused on estimated marginal slopes of prior growth, computed using the *emtrends* function in the *emmeans* package. These slopes represent the expected change in log scale insect abundance per unit increase in prior-year growth within each growth stage $\times$ canopy position combination. Effect sizes were reported as incidence rate ratios (IRRs) by exponentiating the slope estimates and were additionally rescaled to represent the effect of a one-standard-deviation increase in prior growth, for ease of interpretation.

RESULTS

Scale insect abundance varied significantly with canopy position and internode age for three of the four taxa examined (Figure 1; H1). For *Lepidosaphes*, abundance was highest in the upper canopy on the current year leader (internode 1) and declined progressively with both increasing internode age and lower canopy position; the upper canopy current year leader supported significantly more scales than any other combination (Figure 1b). *Parthenolecanium* showed the opposite age pattern, with the highest densities found on the prior year's internode (internode 2) rather than on the current year leader; this peak occurred specifically in the upper canopy, where mean abundance was an order of magnitude greater than at any other position $\times$ internode combination (Figure 1d). Crawler abundance was highest on the two youngest internodes (1 and 2) of upper canopy branches and declined sharply across both older internodes and lower canopy positions (Figure 1c). In contrast, *Phenacoccus* showed no significant pairwise difference among any of the canopy position $\times$ internode age combinations (Figure 1a). A significant main effect of internode age was nonetheless detected in the omnibus test ($\chi^2 = 16.95$; $df = 3$; $p < 0.001$; Table 1), but the high variance with this taxon's low

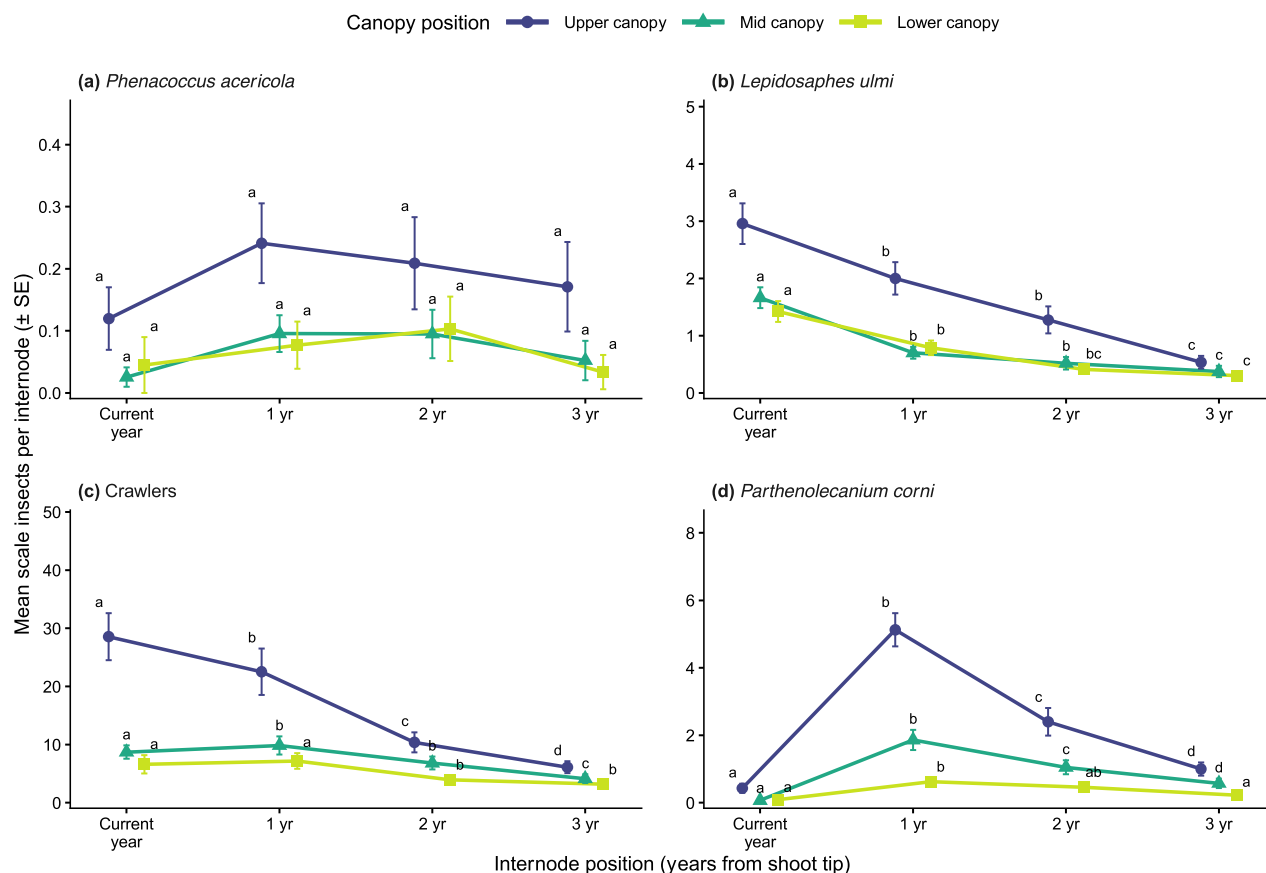


FIGURE 1 Mean scale insect abundance (counts per internode) on sugar maple (*Acer saccharum*) as a function of canopy position (upper, middle, lower crown) and internode position (years from shoot tip). Samples were collected from 33 trees in a recently harvested hardwood stand in south-central Ontario, Canada, during August–September 2015. Means are shown $\pm$ 1 SE. Letters indicate Tukey-adjusted pairwise comparisons among all canopy $\times$ internode age combinations within each species panel; means sharing a letter differ significantly ($p < 0.05$). Sample sizes: *Phenacoccus acericola* ($n = 181$), *Lepidosaphes ulmi* ($n = 1970$), crawlers ($n = 17,151$), *Parthenolecanium corni* ($n = 1997$).

overall abundance precluded identification of specific significantly distinct groupings after Tukey correction.

Canopy stratum and tree size interacted to shape scale insect distributions, although tree size on its own was not a significant predictor for any species (H2). Pairwise comparisons of diameter at breast height (dbh) classes averaged across canopy positions were non-significant for every taxon (Tukey-adjusted $p > 0.4$ in all cases), indicating no detectable main effect of tree size on scale insect abundance. The parallel model treating dbh as a continuous predictor revealed a significant quadratic relationship between tree size and abundance for two of the four taxa (crawlers, $p = 0.006$; *Parthenolecanium*, $p < 0.001$) with predicted abundance peaking at intermediate dbh (approximately 35–45 cm). Canopy position remained a significant predictor for all four taxa in this model ($p \leq 0.014$ in all cases; Figure S2).

Patterns instead emerged within canopy strata. *Parthenolecanium* was concentrated in the upper canopy and the upper canopy abundance increased with tree size, peaking on branches of large trees (55+ cm dbh; Figure 2c); lower-canopy branches of mid-sized trees (30–55 cm) supported the lowest mean densities. *Lepidosaphes*

was similarly concentrated in the upper canopy, with the upper canopy of small trees (15–30 cm) forming the highest abundance group and differing significantly from lower-canopy combinations (Figure 2b).

Crawler abundance was highest in the upper canopy across all tree size classes, but pairwise comparisons revealed overlap among canopy $\times$ size class combinations (Figure 2d), and no single combination was significantly elevated above all others. For *Phenacoccus*, no position $\times$ tree size combination differed significantly from any other (Figure 2a), despite a significant canopy position main effect in the omnibus test ($\chi^2 = 7.12$; $df = 2$; $p = 0.028$; Table 1), but individual pairwise contrasts among position $\times$ size class did not survive Tukey adjustment (Figure 2a), likely reflecting the low sample size for this taxon ($n = 181$).

Spearman's rank correlations among scale insect taxa revealed positive associations whose strength depended on whether shared habitat use was accounted for (Table 2; H3). Raw correlations of branch-level scale densities (counts per cm^2 of bark surface) identified significant positive associations between *Phenacoccus* and *Parthenolecanium* ($\rho = 0.34$; $p < 0.001$), *Parthenolecanium* and crawlers

TABLE 1 Effects of canopy position (upper, middle, lower crown) and internode age on scale insect abundance, and effects of canopy position and tree size class (dbh) on branch-level scale insect abundance, in sugar maple (*Acer saccharum*) sampled in a recently harvested hardwood stand in south-central Ontario, Canada, during August–September 2015.

	Model term	Chi-square	df	p value	
<i>Phenacoccus</i>					
Internode level	Canopy position	2.87	2	0.238	
Internode level	Internode	16.95	3	< 0.001	***
Internode level	Canopy position:internode	2.34	6	0.886	
Branch level	Canopy position	7.12	2	0.028	*
Branch level	dbh	0.46	2	0.794	
Branch level	Canopy position:dbh	5.83	4	0.212	
<i>Lepidosaphes</i>					
Internode level	Canopy position	43.27	2	< 0.001	***
Internode level	Internode	240.52	3	< 0.001	***
Internode level	Canopy position:internode	5.36	6	0.498	
Branch level	Canopy position	29.68	2	< 0.001	***
Branch level	dbh	0.02	2	0.988	
Branch level	Canopy position:dbh	3.47	4	0.483	
<i>Parthenolecanium</i>					
Internode level	Canopy position	107.62	2	< 0.001	***
Internode level	Internode	287.79	3	< 0.001	***
Internode level	Canopy position:internode	19.04	6	< 0.01	**
Branch level	Canopy position	87.01	2	< 0.001	***
Branch level	dbh	0.57	2	0.751	
Branch level	Canopy position:dbh	5.20	4	0.267	
<i>Crawlers</i>					
Internode level	Canopy position	79.44	2	< 0.001	***
Internode level	Internode	212.09	3	< 0.001	***
Internode level	Canopy position:internode	28.48	6	< 0.001	***
Branch level	Canopy position	55.29	2	< 0.001	***
Branch level	dbh	1.61	2	0.447	
Branch level	Canopy position:dbh	1.59	4	0.811	

Note: Results are from per-species negative binomial GLMMs fit with glmmTMB, with tree identity and branch identity included as random intercepts; branch identity was included as an additional random intercept in the internode-level models. Reported χ^2 statistics and p -values are Wald tests (type II) for fixed effects. Adult scale insects were recorded for three observed taxa (*Phenacoccus acericola* $n = 181$; *Parthenolecanium corni* $n = 1997$; *Lepidosaphes ulmi* $n = 1970$) and early-instar crawlers not identified to species ($n = 17,151$). Significance: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

($p = 0.21$; $p < 0.001$) and *Phenacoccus* and crawlers ($p = 0.16$; $p < 0.001$); *Lepidosaphes* was not significantly correlated with any other taxon in the raw analysis (all $|p| \leq 0.09$; $p > 0.05$). After accounting for canopy position and tree size class (dbh) by correlating residuals from log-linear model of branch density, most interspecific associations were weak, but several positive habitat-controlled correlations remained (Table 2; upper triangle). The strongest residual association was between *Parthenolecanium* and *Phenacoccus* (Spearman $\rho = 0.254$; $p < 0.001$), followed by crawlers and *Parthenolecanium* ($\rho = 0.213$; $p < 0.001$). Residual correlations were smaller but still significant for crawlers–*Phenacoccus* ($\rho = 0.109$; $p = 0.023$) and *Lepidosaphes*–*Parthenolecanium* ($\rho = 0.095$; $p = 0.047$). In contrast, *Lepidosaphes* showed no detectable residual association with

Phenacoccus ($\rho = 0.040$; $p = 0.409$) or crawlers ($\rho = 0.005$; $p = 0.918$), indicating little co-variation beyond shared canopy position and tree size class for those pairs.

Estimated marginal trends revealed that the relationship between year-over-year change in internode elongation and subsequent scale insect abundance depended on both growth stage and canopy position (H4). During the 3–2 growth stage, prior elongation was positively associated with scale abundance in the upper and mid canopy. A one-standard-deviation increase in prior internode length (SD = 5.62 cm) was associated with a 25% increase in expected scale abundance in the upper canopy (IRR_{SD} = 1.25; $p = 0.003$) and a 40% increase in the mid canopy (IRR_{SD} = 1.40; $p < 0.001$); the lower canopy showed a positive trend of similar magnitude that did not reach

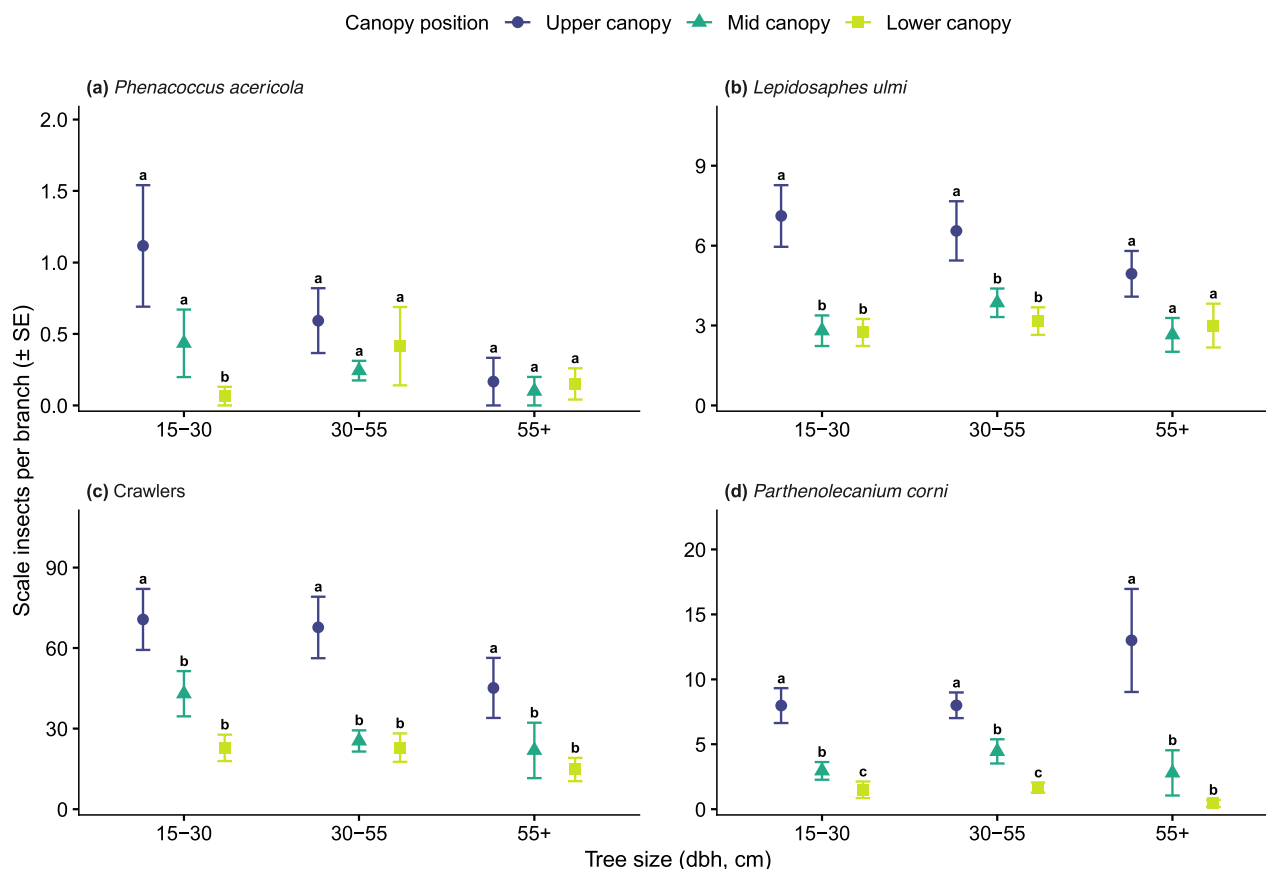


FIGURE 2 Mean scale insect abundance (counts per branch) on sugar maple (*Acer saccharum*) across canopy strata and tree size classes (diameter at breast height; dbh). Branches were collected from the upper, middle and lower canopy of 33 trees in south-central Ontario, Canada, during August–September 2015. Points show the mean count per branch ($\pm$ 1 SE), overlaid on the individual per-branch values (points behind). Within each species panel, letters indicate Tukey-adjusted pairwise comparisons among the three canopy positions within each tree size class; canopy positions that do not share a letter differ significantly ($p < 0.05$). Sample sizes: *Phenacoccus acericola* ($n = 181$), *Lepidosaphes ulmi* ($n = 1970$), crawlers ($n = 17,151$), *Parthenolecanium corni* ($n = 1997$).

TABLE 2 Spearman's rank correlations among scale insect densities and pooled early-instar crawler densities on sugar maple branches.

Species	<i>Phenacoccus</i>	<i>Lepidosaphes</i>	<i>Parthenolecanium</i>	Crawlers
<i>Phenacoccus</i>	—	0.040	0.254 ***	0.109 *
<i>Lepidosaphes</i>	0.032	—	0.095 *	0.005
<i>Parthenolecanium</i>	0.337 ***	0.084	—	0.213 ***
Crawlers	0.159 ***	−0.015	0.209 ***	—

Note: The lower triangle shows raw correlations of branch-level scale insect densities (counts summed across the four annual internodes within each branch, divided by total lateral branch surface area). The upper triangle shows Spearman correlations of linear-model residuals from log-transformed density models fit with canopy position and tree size class as categorical predictors, representing co-variation among taxa after accounting for shared habitat associations. $n = 438$ branches. Significance: * $p < 0.05$; *** $p < 0.001$.

significance ($p = 0.072$). During the 2–1 growth stage, prior elongation was positively associated with scale abundance only in the lower canopy ($IRR_{SD} = 1.32$; $p = 0.022$), whereas no significant relationship was observed in the upper or mid canopy. No significant effects of prior elongation were detected during the 4–3 growth stage in any canopy position.

Prior-year radial growth (internode width) was negatively associated with subsequent scale insect abundance primarily in the upper

and mid canopy. After removing five observations with implausible year-over-year width changes ($\Delta\text{width} > 20$ mm) identified as data entry errors, significant negative relationships were detected in the upper canopy during the 2–1 and 3–2 growth stages ($IRR_{SD} = 0.74$; $p < 0.001$ and $IRR_{SD} = 0.80$; $p = 0.019$, respectively; Figure 3b), and in the mid canopy during both the same two growth stages ($IRR_{SD} = 0.78$; $p = 0.021$ and $IRR_{SD} = 0.79$; $p = 0.039$). No significant width–abundance relationships were detected in the lower canopy

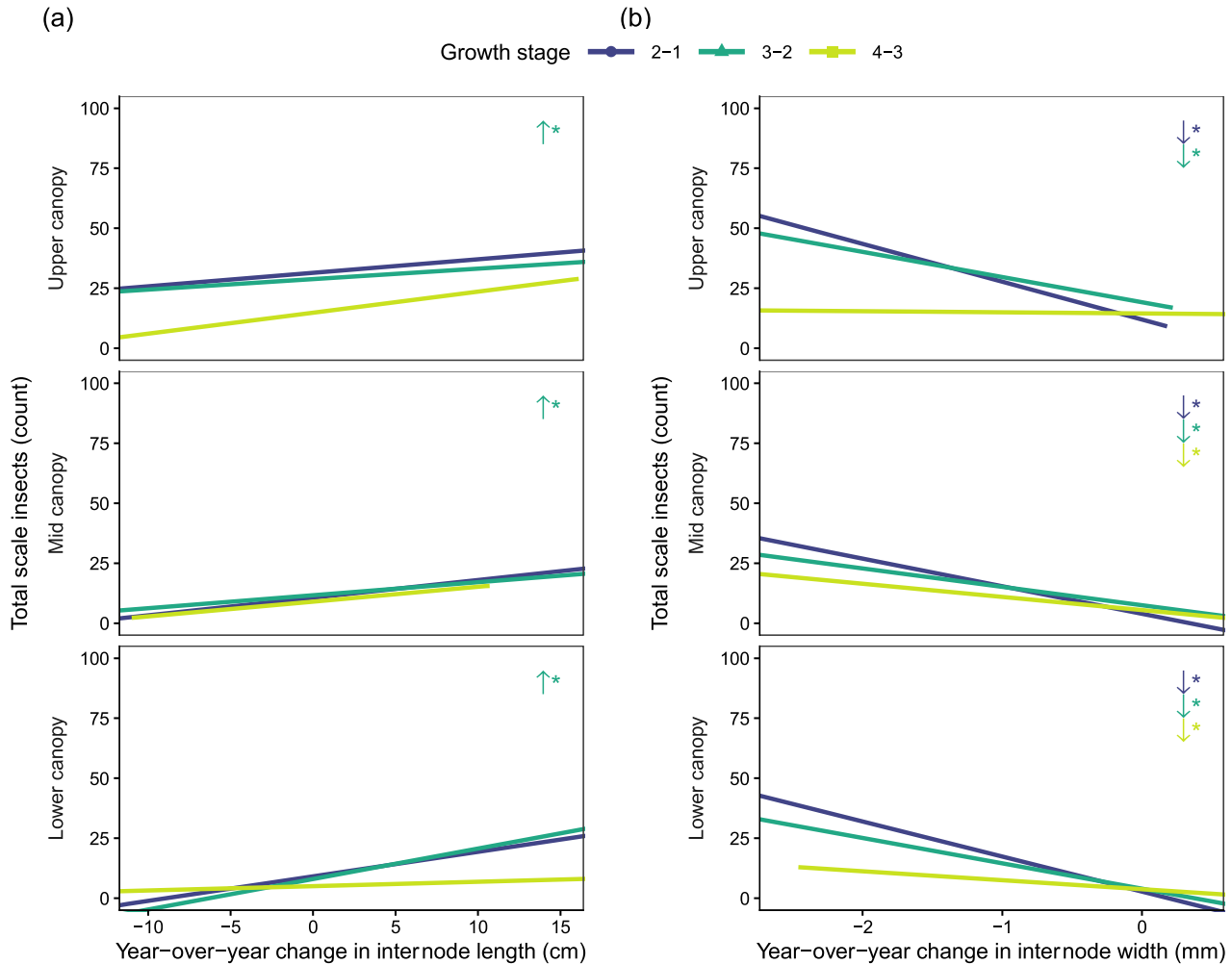


FIGURE 3 Effects of year-over-year change in internode growth on subsequent scale insect abundance across canopy strata on sugar maple (*Acer saccharum*). Data are from 33 trees sampled in a recently harvested hardwood stand in south-central Ontario, Canada, during August–September 2015. Points show observed total scale insect counts per internode; lines show fitted relationships from negative binomial mixed-effects models within each growth stage $\times$ canopy stratum combination. Directional arrows indicate statistically significant estimated marginal slopes ($\uparrow$ positive association; $\downarrow$ negative association; $p < 0.05$). Total scale counts across all taxa were used as the response variable. Sample sizes: *Phenacoccus acericola* ($n = 181$), *Lepidosaphes ulmi* ($n = 1970$), crawlers ($n = 17,151$), *Parthenolecanium corni* ($n = 1997$).

during any growth stage, nor in the upper or mid canopy at the 4–3 stage. Taken together, prior-year elongation and radial growth exhibited contrasting effects on subsequent scale insect abundance, with radial growth effects concentrated in the upper and mid canopy at the two most recent growth stages.

DISCUSSION

Scale insects showed pronounced habitat associations within and among trees that varied among species, with all taxa reaching highest abundance in the upper crown but differing in their associations with internode age and tree size. Co-occurrence patterns among taxa were largely explained by shared habitat use; *Phenacoccus* and *Parthenolecanium* remained strongly correlated even after accounting for canopy position and tree size, whereas *Lepidosaphes* showed only weak

associations with the other taxa once habitat was controlled for. Together, these patterns suggest that scale insect distributions within trees are driven by a combination of host structural traits, physiological gradients related to growth and access to resources and biological limitations on dispersal.

Previous studies of scale insect distribution in forest canopies have taken place in New Zealand with a focus on *Nothofagus* trees (Kelly, 1990; Wardhaugh et al., 2006). The highest densities of *Ultra-coelostoma assimile* Maskell scale there occurred on medium-sized tree trunks (Kelly, 1990) and on branches and trunks of medium-sized *Nothofagus* trees (10–30 cm dbh) (Wardhaugh et al., 2006). In the present study *Phenacoccus* and immature crawlers showed a pattern qualitatively similar to these results, having somewhat higher abundance on trees between 15 and 30 cm diameter at breast height (dbh) (Figure 2a), though pairwise differences between tree size classes were not statistically significant in our analysis. However, the patterns

diverged in *Parthenolecanium*, which showed a higher abundance on the upper canopy of large trees (Figure 2c). Our results thus indicate that a range of patterns may exist in scale insect abundance relative to tree size, and that interactions with canopy position appear more important than tree size effects on their own.

Patterns of scale distribution were more consistent for canopy stratification than tree size: All scale species examined were most abundant in the upper canopy. This is consistent with a prior study on *Ultracoelostoma* scale in *Nothofagus* trees that also found densities of scale insects five times greater on branches than main stems (Wardhaugh et al., 2006). Existing evidence thus suggests a pervasive pattern of increased scale insect abundance in the upper canopies of trees. Several factors likely contribute to this pattern. Scale insects require access to areas where phloem cells are close to the surface, but they also require crevices within the bark to provide shelter and protection. Bark thickness is almost certainly a key driver of scale distribution (Jactel et al., 1996; Nicolai, 1986; Wardhaugh et al., 2006). Nicolai (1986) observed that crawlers mainly colonized newly created fissures in tree bark, suggesting the importance of protected microenvironments for this life-history stage. Jactel et al. (1996) described the vertical distribution of the margarodid scale insect, *Matsucoccus feytaudi* Ducas, on the trunks of maritime pine (*Pinus pinaster* Aiton): Scale insect abundance was affected by bark thickness, with very few insects colonizing either the lower trunk where bark was thick or the top of the trunk where bark was thin and smooth. Young sugar maple trees have relatively smooth bark compared to older trees with bark fissures (Gauthier & Guillemette, 2018), and the more developed bark architecture of mature trees may provide some structural shelter for settling crawlers, contributing to the somewhat higher numerical abundance of scales we observed on mature trees.

Plant chemistry is another factor likely to influence the heterogeneity of scale species within tree canopies; in particular, the chemical composition of host tree phloem has been shown to influence sap-sucking insect distribution (Sousa-Souto et al., 2018). Nitrogen is a key limiting resource for sap-feeding insects (Douglas, 2006; Mattson, 1980), which generally show pronounced increases in survival and fecundity with increases in host plant nitrogen status (Huberty & Denno, 2006). Thomas et al. (2010) found that leaf nitrogen content of sugar maple was highest among trees between 5 and 10 cm dbh, with a decline in larger trees. Prior studies (mainly of cultivated fruit trees) have also documented a close correlation between foliar N and the concentrations of N compounds in phloem (Youssefi et al., 2000). Secondary plant metabolites, such as alkaloids and glycosides, used by plants to deter sap-sucking pests (Barbehenn et al., 2005; Barbehenn & Kochmanski, 2013; Stevenson et al., 1996) are unevenly distributed within individual trees (Sousa-Souto et al., 2018) and have been suggested to influence within-tree scale distribution (Fry et al., 2009; Wardhaugh et al., 2006). Here we found that trees in the 15–30 cm dbh size category, which are expected to show relatively high N concentrations in the phloem, also showed numerically elevated abundances of *Phenacoccus* and immature crawlers (Figure 2a,d), although these tree size differences were not statistically significant when averaged across canopy positions. However,

this was not the case for *Parthenolecanium*, which was numerically more abundant in larger tree categories, particularly within the upper canopy (Figure 2c).

Importantly, our data further indicate that the two dimensions of annual branch growth had opposite effects on subsequent scale densities. The contrasting effects of growth dimensions observed here provide insight into the mechanisms underlying scale insect distribution within host trees. Increased internode elongation was consistently associated with higher scale insect abundance (Figure 3a), suggesting that rapidly elongating shoots provide enhanced phloem flux or higher concentrations of soluble nutrients available to sap-feeding insects (Douglas, 2006; Mattson, 1980; Price, 1991). In contrast, increased radial growth was negatively associated with scale abundance in the upper and mid canopy (Figure 3b), consistent with the interpretation that thicker annual bark growth limits crawler access to underlying phloem tissues (Gauthier & Guillemette, 2018; Wardhaugh, 2014). Notably, these effects were strongest where scale abundance was highest, suggesting that bark-mediated phloem accessibility is a particularly important constraint in the most heavily colonized parts of the crown. Together, these contrasting growth-related patterns suggest that scale insect distributions are shaped by fine-scale variation in shoot architecture and resource flow, rather than overall host vigour alone.

Forest organisms that live within tree canopies experience distinct microclimates depending on canopy stratum (Nakamura et al., 2017). For organisms with limited dispersal capacity, such as scale insects, these microclimatic differences can have lasting impacts. Scale insects are characterized by a brief dispersal phase limited to the crawler stage, after which they become effectively sessile for the remainder of their development (Magsig-Castillo et al., 2010; Matile-Ferrero, 1997; Wright et al., 2025). Consequently, spatial patterns observed on branch internodes likely reflect conditions experienced during initial settlement. Thus, settlement on growth segments with favourable structural and nutritional conditions is likely to have persistent effects on adult distributions across subsequent years. The vertical stratification patterns of various arthropod groups have been examined in northern hardwood forests (Patankar et al., 2011; Thomas et al., 2010; Vance et al., 2007) with most herbivore groups found to be at highest densities in the lower- or mid-canopies (Thomas et al., 2010). In contrast, despite an overall underrepresentation of upper canopy specialists in this system, the most abundant herbivorous arthropod, the maple spindle gall mite (*Vasates aceriscrumena* Riley), showed peak abundance in the upper canopy (Patankar et al., 2011; Thomas et al., 2010). Gall-forming arthropods such as eriophyoid mites are comparable in many respects to scale insects in that they are essentially sessile following early juvenile dispersal and later are highly protected from desiccation. In contrast, sap-feeding insects that are not strongly protected from desiccation have not shown increased abundance in the upper canopy. Prior studies on aphids, an insect closely related to scales but lacking adaptations to reduce desiccation (Matile-Ferrero, 1997), have found peak densities in the middle or lower portion of the host plant (Fernandes et al., 2012), or in some cases, no strong vertical stratification (McClure, 1982; Platková et al., 2020). Similarly, a study of adelgids

on *Pinus* species found greater abundance in the lower part of the crown (McClure, 1982).

Distinct vertical canopy stratification in arthropod distributions is particularly evident in tropical forests, leading some authors to suggest a lack of such patterns in temperate forest systems (Fowler, 1985; Lowman & Wittman, 1996). More recent literature, however, has clearly documented vertical stratification of arthropods in temperate forests, spanning multiple feeding guilds including herbivores, predators and detritivores (Hakimara & Despland, 2025; Maguire et al., 2014; Seifert et al., 2020; Thomas et al., 2010). The present study supports the conclusion that scale insects show substantially increased abundance in the upper canopy, but also that the patterns differ somewhat among species, in particular in relation to tree size. Despite the abundance of scale insects in temperate hardwood forests and their potential to provide spatially structured carbohydrate resources to higher-trophic consumers, the within-tree distribution of these insects in northern temperate systems has received limited attention. By documenting how scale insect abundance varies with both canopy stratum and tree size, and how year-over-year changes in branch growth shape subsequent scale densities, our findings advance understanding of how canopy heterogeneity structures sap-feeding insect communities and the spatial availability of honeydew-based resource subsidies. Due to their importance in providing resources in the form of carbohydrate-rich honeydew to a variety of other organisms, notably bees and other nectar feeders such as adult parasitic wasps and lepidopterans (Moller & Tilley, 1989; Ewers, 2002; dos Santos et al., 2019; Requier & Leonhardt, 2020), the vertical and age-related patterns described here are likely to have important effects on broader community-level patterns.

AUTHOR CONTRIBUTIONS

Susan M. Frye: Conceptualization; methodology; formal analysis; writing – original draft; project administration; visualization; investigation; software. **Sandy M. Smith:** Funding acquisition; writing – review and editing; supervision; resources. **Sean C. Thomas:** Conceptualization; methodology; funding acquisition; writing – review and editing; supervision; investigation.

ACKNOWLEDGEMENTS

We thank I. Kennedy and T. Ung for field assistance, P. Vomero and E. Chang for sample processing, L. Seehausen for statistical help, and the Haliburton Forest and Wild Life Reserve for access to field sites. This work was supported by the Natural Sciences and Engineering Research Council of Canada (NSERC) Discovery grants to SMS and SCT.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The data supporting the results of this study are archived in Borealis, the Canadian Dataverse Repository (<https://doi.org/10.5683/SP4/HAVQQK>).

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REFERENCES

- Ampoorter, E., Barbaro, L., Jactel, H., Baeten, L., Boberg, J., Carnol, M. et al. (2020) Tree diversity is key for promoting the diversity and abundance of forest-associated taxa in Europe. *Oikos*, 129, 133–146. Available from: <https://doi.org/10.1111/oik.06290>
- Awmack, C.S. & Leather, S.R. (2002) Host plant quality and fecundity in herbivorous insects. *Annual Review of Entomology*, 47, 817–844. Available from: <https://doi.org/10.1146/annurev.ento.47.091201.145300>
- Barbehenn, R., Cheek, S., Gasperut, A., Lister, E. & Maben, R. (2005) Phenolic compounds in red oak and sugar maple leaves have prooxidant activities in the midgut fluids of *Malacosoma disstria* and *Orgyia leucostigma* caterpillars. *Journal of Chemical Ecology*, 31, 969–988. Available from: <https://doi.org/10.1007/s10886-005-4242-4>
- Barbehenn, R.V. & Kochmanski, J. (2013) Searching for synergism: effects of combinations of phenolic compounds and other toxins on oxidative stress in *Lymantria dispar* caterpillars. *Chemoecology*, 23, 219–231. Available from: <https://doi.org/10.1007/s00049-013-0136-z>
- Birtele, D. & Hardersen, S. (2012) Analysis of vertical stratification of Syrphidae (Diptera) in an oak-hornbeam forest in northern Italy. *Ecological Research*, 27, 755–763. Available from: <https://doi.org/10.1007/s11284-012-0948-2>
- Boege, K., Barton, K.E. & Dirzo, R. (2011) Size- and age-related changes in tree structure and function. In: Meinzer, F.C., Lachenbruch, B. & Dawson, T.E. (Eds.) *Size- and age-related changes in tree structure and function*. Tree physiology, Vol. 4. Dordrecht: Springer, pp. 193–214. Available from: https://doi.org/10.1007/978-94-007-1242-3_7
- Brooks, M.E., Kristensen, K., van Benthem, K.J., Magnusson, A., Berg, C.W., Nielsen, A. et al. (2017) glmmTMB balances speed and flexibility among packages for zero-inflated generalized linear mixed models. *R Journal*, 9, 378–400. Available from: <https://doi.org/10.32614/RJ-2017-066>
- Chilima, C.Z. & Leather, S.R. (2001) Within-tree and seasonal distribution of the pine woolly aphid *Pineus boemeri* on *Pinus kesiya* trees. *Agricultural and Forest Entomology*, 3, 139–145. Available from: <https://doi.org/10.1046/j.1461-9563.2001.00098.x>
- Corff, J.L. & Marquis, R.J. (1999) Differences between understorey and canopy in herbivore community composition and leaf quality for two oak species in Missouri. *Ecological Entomology*, 24, 46–58. Available from: <https://doi.org/10.1046/j.1365-2311.1999.00174.x>
- Crozier, L.R. (1981) Beech honeydew: forest produce. *New Zealand Journal of Forestry*, 26, 200–209.
- Dale, A.G. & Frank, S.D. (2017) Warming and drought combine to increase pest insect fitness on urban trees. *PLoS One*, 12, e0173844. Available from: <https://doi.org/10.1371/journal.pone.0173844>
- dos Santos, C.F., Halinski, R., dos Santos, P.D.S., Almeida, E.A.B. & Blochtein, B. (2019) Looking beyond the flowers: associations of stingless bees with sap-sucking insects. *The Science of Nature*, 106, 12. Available from: <https://doi.org/10.1007/s00114-019-1608-y>
- Douglas, A.E. (2006) Phloem-sap feeding by animals: problems and solutions. *Journal of Experimental Botany*, 57, 747–754. Available from: <https://doi.org/10.1093/jxb/erj067>
- Eubanks, M.D. & Styrsky, J.D. (2005) Effects of plant feeding on the performance of omnivorous ‘predators’. In: Wäckers, F.L., van Rijn, P.C.-J. & Bruin, J. (Eds.) *Plant-provided food for carnivorous insects: a protective mutualism and its applications*. Cambridge, UK: Cambridge University Press, pp. 148–177. Available from: <https://doi.org/10.1017/CBO9780511542220.007>

- Ewers, R.M. (2002) The influence of honeydew on arthropod community composition in a New Zealand beech forest. *New Zealand Journal of Ecology*, 26, 23–29.
- Fernandes, F.S., Ramalho, F.S., Malaquias, J.B., Junior, J.L.N., Correia, E.T. & Zancunio, J.C. (2012) Within-plant distribution of cotton aphid (Hemiptera: Aphididae) in cotton cultivars with colored fibers. *Anais da Academia Brasileira de Ciencias*, 84, 707–721. Available from: <https://doi.org/10.1590/s0001-37652012005000040>
- Fowler, S.V. (1985) Differences in insect species richness and faunal composition of birch seedlings, saplings and trees: the importance of plant architecture. *Ecological Entomology*, 10, 159–169. Available from: <https://doi.org/10.1111/j.1365-2311.1985.tb00545.x>
- Fry, H.R.C., Quiring, D.T., Ryall, K.L. & Dixon, P.L. (2009) Influence of intra-tree variation in phenology and oviposition site on the distribution and performance of *Ennomos subsignaria* on mature sycamore maple. *Ecological Entomology*, 34, 394–405. Available from: <https://doi.org/10.1111/j.1365-2311.2009.01091.x>
- Gauthier, M.-M. & Guillemette, F. (2018) Bark type reflects growth potential of yellow birch and sugar maple at the northern limit of their range. *Plant Ecology*, 219, 381–390. Available from: <https://doi.org/10.1007/s11258-018-0802-x>
- Gaze, P.D. & Clout, M.N. (1983) Honeydew and its importance to birds in beech forests of South Island, New Zealand. *New Zealand Journal of Ecology*, 6, 33–37.
- Grant, W.D. & Beggs, J.R. (1989) Carbohydrate analysis of beech honeydew. *New Zealand Journal of Zoology*, 16, 283–288. Available from: <https://doi.org/10.1080/03014223.1989.10422893>
- Hakimara, M. & Despland, E. (2025) Vertical stratification of leaf physical traits exerts bottom-up pressures on insect herbivory in a sugar maple temperate forest. *Insect Conservation and Diversity*, 18, 1–15. Available from: <https://doi.org/10.1111/icad.12777>
- Hanks, L.M. & Denno, R.F. (1993) Natural enemies and plant water relations influence the distribution of an armored scale insect. *Ecology*, 74, 1081–1091. Available from: <https://doi.org/10.2307/1940478>
- Harrell, F.E.J. (2024) Hmisc: Harrell Miscellaneous. R Package Version 5.1. <https://CRAN.R-project.org/package=Hmisc>
- Hothorn, T., Bretz, F. & Westfall, P. (2008) Simultaneous inference in general parametric models. *Biometrical Journal*, 50, 346–363. Available from: <https://doi.org/10.1002/bimj.200810425>
- Huberty, A.F. & Denno, R.F. (2006) Consequences of nitrogen and phosphorus limitation for the performance of two planthoppers with divergent life-history strategies. *Oecologia*, 149, 444–455. Available from: <https://doi.org/10.1007/s00442-006-0462-8>
- Jactel, H., Perthuisot, N., Menassieu, P., Raise, G. & Burban, C. (1996) A sampling design for within-tree larval populations of the maritime pine scale, *Matsucoccus feytaudi* Duc. (Homoptera: Margarodidae), and the relationship between larval population estimates and male catch in pheromone traps. *The Canadian Entomologist*, 128, 1143–1156. Available from: <https://doi.org/10.4039/Ent1281143-6>
- Kelly, D. (1990) Honeydew density in mixed *Nothofagus* forest, Westland, New Zealand. *New Zealand Journal of Botany*, 28, 53–58. Available from: <https://doi.org/10.1080/0028825X.1990.10412343>
- Khaladi, O., Bouderbala, A., Mahdjoubi, D., Benada, M. & Boumaaza, B. (2024) Exploring scale insects biodiversity in orange orchards: insights from Guelma province, Algeria. *Archives of Phytopathology and Plant Protection*, 57, 555–567. Available from: <https://doi.org/10.1080/03235408.2024.2375042>
- Lenth, R.V. (2024) Emmeans: estimated marginal means, aka least-squares means. R Package Version 1.10. Available Fromm: <https://CRAN.R-project.org/package=emmeans>
- Linghu, W., Lu, Z., Wang, Y. & Gao, G. (2023) The effects of globose scale (*Sphaerolecanium prunastri*) infestation on the growth of wild apricot (*Prunus armeniaca*) trees. *Forests*, 14, 2032. Available from: <https://doi.org/10.3390/f14102032>
- Lowman, M.D. & Wittman, P.K. (1996) Forest canopies: methods, hypotheses, and future directions. *Annual Review of Ecology and Systematics*, 27, 55–81. Available from: <https://doi.org/10.1146/annurev.ecolsys.27.1.55>
- Magsig-Castillo, J., Morse, J.G., Walker, G.P., Bi, J.L., Rugman-Jones, P.F. & Stouthamer, R. (2010) Phoretic dispersal of armored scale crawlers (Hemiptera: Diaspididae). *Journal of Economic Entomology*, 103, 1172–1179. Available from: <https://doi.org/10.1603/EC10030>
- Maguire, D.Y., Robert, K., Brochu, K., Larrivée, M., Buddle, C.M. & Wheeler, T.A. (2014) Vertical stratification of beetles (coleoptera) and flies (Diptera) in temperate forest canopies. *Environmental Entomology*, 43, 9–17. Available from: <https://doi.org/10.1603/EN13056>
- Matile-Ferrero, D. (1997) Morphology: the adult female. In: Ben-Dov, Y. & Hodgson, C.J. (Eds.) *Soft scale insects: their biology, natural enemies and control*. Amsterdam: Elsevier, pp. 5–21.
- Mattson, W.J. (1980) Herbivory in relation to plant nitrogen content. *Annual Review of Ecology and Systematics*, 11, 119–161. Available from: <https://doi.org/10.1146/annurev.es.11.110180.001003>
- McClure, M.S. (1982) Distribution and damage of two *Pineus* species (Homoptera: Adelgidae) on red pine in New England. *Annals of the Entomological Society of America*, 75, 150–157. Available from: <https://doi.org/10.1093/aesa/75.2.150>
- Miller, D.R. & Davidson, J.A. (2005) *Armored scale insect pests of trees and shrubs (Hemiptera: Diaspididae)*. Ithaca, NY: Cornell University Press.
- Moller, H. & Tilley, J.A.V. (1989) Beech honeydew: seasonal variation and use by wasps, honey bees, and other insects. *New Zealand Journal of Zoology*, 16, 289–302. Available from: <https://doi.org/10.1080/03014223.1989.10422894>
- Morales, M.G., Denno, B.D., Miller, D.R., Miller, G.L. & Ben-Dov, Y. (2016) ScaleNet: a literature-based model of scale insect biology and systematics. *Database*, 2016, bav118. Available from: <https://doi.org/10.1093/database/bav118>
- Nakamura, A., Kitching, R.L., Cao, M., Creedy, T.J., Fayle, T.M., Freiberg, M. et al. (2017) Forests and their canopies: achievements and horizons in canopy science. *Trends in Ecology & Evolution*, 32, 438–451. Available from: <https://doi.org/10.1016/j.tree.2017.02.020>
- Nicolai, V. (1986) The bark of trees: thermal properties, microclimate and fauna. *Oecologia*, 69, 148–160. Available from: <https://doi.org/10.1007/BF00399052>
- Nock, C.A., Caspersen, J.P. & Thomas, S.C. (2008) Large ontogenetic declines in intra-crown leaf area index in two temperate deciduous tree species. *Ecology*, 89, 744–753. Available from: <https://doi.org/10.1890/07-0531.1>
- OMNR (Ontario Ministry of Natural Resources). (2015) *Forest management guide to silviculture in the Great Lakes-St. Lawrence and Boreal forests of Ontario (publication No. 109091_U)*. Toronto, Canada: Queen's Printer for Ontario.
- Patankar, R., Fuller, M.M., Smith, S.M. & Thomas, S.C. (2014) The distribution of a host-specific canopy parasite is linked with local species diversity in a northern temperate forest. *Journal of Vegetation Science*, 25, 1015–1023. Available from: <https://doi.org/10.1111/jvs.12104>
- Patankar, R., Thomas, S.C. & Smith, S.M. (2011) A gall-inducing arthropod drives declines in canopy tree photosynthesis. *Oecologia*, 167, 701–709. Available from: <https://doi.org/10.1007/s00442-011-2019-8>
- Phillipson, J. & Thompson, D.J. (1983) Phenology and intensity of phyllophage attack on *Fagus sylvatica* in Wytham woods, Oxford. *Ecological Entomology*, 8, 315–330. Available from: <https://doi.org/10.1111/j.1365-2311.1983.tb00512.x>
- Platková, H., Pyszkó, P., D'Acier, A.C., Jousselin, E. & Drozd, P. (2020) Spatial distribution of aphids in the canopy of a temperate forest: where can they be found? *Agricultural and Forest Entomology*, 22, 379–389. Available from: <https://doi.org/10.1111/afe.12393>
- Price, P.W. (1991) The plant vigor hypothesis and herbivore attack. *Oikos*, 62, 244–251. Available from: <https://doi.org/10.2307/3545270>
- R Core Team. (2025) *R: a language and environment for statistical computing*. Vienna, Austria: R Foundation for Statistical Computing. Available from: <https://www.R-project.org/>

- Requier, F. & Leonhardt, S.D. (2020) Beyond flowers; including non-floral resources in bee conservation schemes. *Journal of Insect Conservation*, 24, 5–16. Available from: <https://doi.org/10.1007/s10841-019-00206-1>
- Schowalter, T.D. (1989) Canopy arthropod community structure and herbivory in old-growth and regenerating forests in western Oregon. *Canadian Journal of Forest Research*, 19, 318–322. Available from: <https://doi.org/10.1139/x89-047>
- Seifert, C.L., Lamarre, G.P.A., Volf, M., Jorge, L.R., Miller, S.E., Wagner, D.L. et al. (2020) Vertical stratification of a temperate forest caterpillar community in eastern North America. *Oecologia*, 192, 501–514. Available from: <https://doi.org/10.1007/s00442-019-04584-w>
- Šigut, M., Šigutová, H., Šipoš, J., Pysko, P., Kotásková, N. & Drozd, P. (2018) Vertical canopy gradient shaping the stratification of leaf-chewer–parasitoid interactions in a temperate forest. *Ecology and Evolution*, 8, 7297–7311. Available from: <https://doi.org/10.1002/ece3.4194>
- Smith, J.P. & Schowalter, T.D. (2001) Aphid-induced reduction of shoot and root growth in Douglas-fir seedlings. *Ecological Entomology*, 26, 411–416. Available from: <https://doi.org/10.1046/j.1365-2311.2001.00335.x>
- Smith, S.M., Islam, N. & Bellocq, M.I. (2012) Effects of single-tree selection harvesting on hymenopteran and saproxylic insect assemblages in the canopy and understory of northern temperate forests. *Journal of Forestry Research*, 23, 275–284. Available from: <https://doi.org/10.1007/s11676-012-0253-5>
- Sousa-Souto, L., Bocchiglieri, A., Dias, D.d.M., Ferreira, A.S. & de Lima Filho, J.P. (2018) Changes in leaf chlorophyll content associated with flowering and its role in the diversity of phytophagous insects in a tree species from a semiarid caatinga. *PeerJ*, 6, e5059. Available from: <https://doi.org/10.7717/peerj.5059>
- Speight, M.R., Hails, R.S., Gilbert, M. & Foggo, A. (1998) Horse chestnut scale (*Pulvinaria regalis*) (Homoptera: Coccidae) and urban host tree environment. *Ecology*, 79, 1503–1513.
- Stevenson, P.C., Kimmins, F.M., Grayer, R.J. & Raveendranath, S. (1996) Schaftosides from rice phloem as feeding inhibitors and resistance factors to brown planthoppers, *Nilaparvata lugens*. *Entomologia Experimentalis et Applicata*, 80, 246–249. Available from: <https://doi.org/10.1111/j.1570-7458.1996.tb00928.x>
- Thomas, S.C. (2010) Photosynthetic capacity peaks at intermediate size in temperate deciduous trees. *Tree Physiology*, 30, 555–573. Available from: <https://doi.org/10.1093/treephys/tpq005>
- Thomas, S.C., Sztaba, A.J. & Smith, S.M. (2010) Herbivory patterns in mature sugar maple: variation with vertical canopy strata and tree ontogeny. *Ecological Entomology*, 35, 1–8. Available from: <https://doi.org/10.1111/j.1365-2311.2009.01133.x>
- Vance, C.C., Smith, S.M., Malcolm, J.R., Huber, J. & Bellocq, M.I. (2007) Differences between forest type and vertical strata in the diversity and composition of hymenopteran families and mymarid genera in northeastern temperate forests. *Environmental Entomology*, 36, 1073–1083. Available from: <https://doi.org/10.1093/ee/36.5.1073>
- Wardhaugh, C.W. (2014) The spatial and temporal distributions of arthropods in forest canopies: uniting disparate patterns with hypotheses for specialisation. *Biological Reviews*, 89, 1021–1041. Available from: <https://doi.org/10.1111/brv.12094>
- Wardhaugh, C.W., Blakely, T.J., Greig, H., Morris, P.D., Barnden, A., Rickard, S. et al. (2006) Vertical stratification in the spatial distribution of the beech scale insect (*Ultracoelostoma assimile*) in *Nothofagus* tree canopies in New Zealand. *Ecological Entomology*, 31, 185–195. Available from: <https://doi.org/10.1111/j.0307-6946.2006.00778.x>
- Wardhaugh, C.W. & Didham, R.K. (2005) Density-dependent effects on the reproductive fitness of the New Zealand beech scale insect (*Ultracoelostoma assimile*) across multiple spatial scales. *Ecological Entomology*, 30, 733–738. Available from: <https://doi.org/10.1111/j.0307-6946.2005.00739.x>
- Wardhaugh, C.W. & Didham, R.K. (2006) Establishment success of sooty beech scale insects, *Ultracoelostoma* sp., on different host tree species in New Zealand. *Journal of Insect Science*, 6, 29. Available from: https://doi.org/10.1673/2006_06_29.1
- Wharton, E.H. (1984) Predicting diameter at breast height from stump diameters for northeastern tree species. Research note NE-322. Broomall, PA: U.S. Department of Agriculture, Forest Service, Northeastern Forest Experiment Station, p. 4. Available from: <https://doi.org/10.2737/NE-RN-322>
- Wright, E.R., Long, D.E., McAndrew, K.M., Paul, T.G., Sabet, A., Riggins, I.O. et al. (2025) Potential crawling distance for an invading urban tree pest: implications for settling decisions and between-tree movement. *Journal of Insect Behavior*, 38, 8. Available from: <https://doi.org/10.1007/s10905-025-09872-3>
- Xing, S., Leahy, L., Ashton, L.A., Kitching, R.L., Bonebrake, T.C. & Scheffers, B.R. (2023) Ecological patterns and processes in the vertical dimension of terrestrial ecosystems. *Journal of Animal Ecology*, 92, 538–551. Available from: <https://doi.org/10.1111/1365-2656.13881>
- Youssefi, F., Brown, P.H. & Weinbaum, S.A. (2000) Relationship between tree nitrogen status, xylem and phloem sap amino acid concentrations, and apparent soil nitrogen uptake by almond trees (*Prunus dulcis*). *Journal of Horticultural Science and Biotechnology*, 75, 62–68. Available from: <https://doi.org/10.1080/14620316.2000.11511201>

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

Figure S1. Distribution of stem diameter at breast height (diameter at breast height; dbh) for the 33 sugar maples (*Acer saccharum*) sampled from the upper, middle and lower canopy in south-central Ontario, Canada, during August–September 2015. Bars (5-cm bins) are coloured by size class, and dashed vertical lines at 15, 30 and 55 cm mark the class breaks; the number of trees in each class is given above it. The <15 cm class ($n = 4$) is included in the continuous-dbh analysis (Figure S2) but not in the categorical size class models (Figure 2), which use the 15–30 ($n = 9$), 30–55 ($n = 16$) and 55+ ($n = 4$) cm classes.

Figure S2. Scale insect abundance (counts per branch) on 33 sugar maple (*Acer saccharum*) trees as a continuous function of tree size (diameter at breast height; dbh), fitted separately for each canopy stratum. Curves are marginal predictions from negative binomial GLMMs that include linear and quadratic dbh terms and an additive canopy-position effect. Tukey-adjusted compact letters at the right end of each curve compare canopy strata (size-adjusted marginal means); strata that do not share a letter differ significantly ($p < 0.05$). Dashed vertical. Lines at 30 and 55 cm mark the tree size class boundaries used in Figure 2. Sample sizes: *Phenacoccus acericola* ($n = 181$), *Lepidosaphes ulmi* ($n = 1970$), crawlers ($n = 17,151$), *Parthenolecanium corni* ($n = 1997$).

How to cite this article: Frye, S.M., Smith, S.M. & Thomas, S.C. (2026) Canopy position and tree size structure the distribution of honeydew-producing scale insects (Hemiptera: Coccoidea) in a temperate hardwood forest. *Ecological Entomology*, 1–12. Available from: <https://doi.org/10.1111/een.70144>